

Wind Power Ranking

India ranks 5th in the world with a total wind power capacity of 10925 MW as of 2009 whereas US is No. 1

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Feature



Atul Gopal (inset) along with a photograph of his drawing room in which he has replaced all the lights with LEDs

generation of about 85 Lumens. We used roughly one for every 10 sq ft. The current lighting is reasonable for most activities, except reading. We deliberately chose a slightly lesser intensity, after spending Rs. 4000 for a new set of LEDs when our electronics to dim the LEDs ended up busting all of them," he says tongue in cheek. The situation has improved now with LED lights coming with 50 per cent more Lumens at a 50 per cent lesser price.

But these failures did not deter Gopal as the savings he encountered with this source of lighting were quite significant. Before LEDs, he used two CFLs that consumed 40 W of power. Using an LED having a power consumption of just 1W, Mr Gopal used 17.5 W in all to light up his 150 sq. ft. drawing room. "I am saving about 22.5 W using LED lights. In terms of rupees, assuming a usage of about 4 hours a day and 300 days a year, the savings are about $1200 * 22.5 = 27$ kWh. My residential electricity rate in Pune is about Rs. 5 per unit. So I end up saving about Rs. 135-150 per year. In percentage terms this is 2 - 3 per cent of my bill, with just one room converted. Also note that I was already using CFLs before this," he said. According to Gopal, the LED lights he has invested in will last him for a good 25-30 years.

Harnessing solar energy

Sealine Society in Bandra (W), Mumbai seems like any other housing society in the neighbourhood, but signboards like 'Water is precious' and 'Conserve Energy'

are out of the ordinary. Only when you walk around the society and have a tour of its terrace, does the gravity of those signboards sink in. Thanks to octogenarian Navin Chandra, Sealine Society has no worries as far as electricity and water shortages are concerned.

A chemical engineer by education, Chandra has worked in the capacity of a CEO of a multinational company

in the past. At an age when most of his peers would rather rest at home, Chandra has taken initiatives that have resulted in his society saving on their monthly electricity bills. One such initiative was to convince his society members to invest in solar panels in 2006.

While taking us to the terrace, Chandra stressed on the importance of such initiatives. "Some years back the cost of solar panels was quite high, but now that price is reducing. Conventional electricity, on the other hand, is rising," he said.

The terrace of the building has a small garden and is surrounded by solar panels, and a solar water heater facility. "We have a total of 24 solar panels. Eight of them are rotary, which means they can move along with the sun such that efficiency is maximized by facing the sun. The energy from these panels is stored in the batteries that are placed in our terrace store room. The DC current is converted to AC by an inverter before this energy is used for lighting up common passageways," he told us. One cannot help noticing the enthusiasm with which Chandra explains the whole setup and answers all queries with a smile.

JNNS MISSION

Jawaharlal Nehru National Solar Mission (JNNSM) is a part of the National Action Plan on Climate Change (NAPCC) which has set a target of achieving 20,000 MW solar power installation by the year 2022. The main idea behind this mission is to bring the price of solar power and grid power on par. The idea of the mission is to build on this and also focus on Research and Development and HRD to develop and strengthen Indian skills thereby enhancing indigenous content to make the mission more sustainable.

The electricity generated by the solar panels which is about 2kW, is enough to light up about 50 lights in the common passageway. According to Chandra, since the common areas are only used during certain part of the day, the use of electricity is spread across the day. This little measure has ensured that the society saves a decent amount of money on its electricity bill.

One question that arised in our minds was, if the solar energy is used for lighting common passageways, then what was stopping members from using it for secondary lighting in their houses. Chandra stated that most of the houses in the society had concealed wiring, and to rewire the house for secondary lighting did not make sense. "Moreover right now we only generate enough solar energy for lighting common passageways. For generating additional energy we will need more solar panels and lack of space on our terrace prevents that," he said.



Navin Chandra standing alongside the solar water heating panels